

SUPPLEMENTAL MATERIAL

Association between ambient fine particulate pollution and hospital admissions for cause specific cardiovascular disease: time series study in 184 major Chinese cities

Table S1: City-specific number of people enrolled in Urban Employee Basic Medical Insurance (UEBMI) in 2017, the number of residents, and the coverage rate based on the UEBMI in 184 Chinese cities.

City	Number of city's residents (thousand)*	Number of people enrolled in the UEBMI in 2017 (thousand)	Coverage rate of the population (%)
An'shan	3645.9	1169.735	32.08
Anyang	5172.8	639.594	12.36
Baicheng	2033.1	238.969	11.75
Baise	3466.8	352.491	10.17
Baiyin	1708.8	267.624	15.66
Baoji	3716.7	578.582	15.57
Baoshan	2506	162.638	6.49
Baotou	2650.3	881.293	33.25
Bayannur	1669.9	258.573	15.48
Bazhong	3283.7	215.918	6.58
Beihai	1539.3	248.703	16.16
Bengbu	3164	518.759	16.40
Benxi	1709.5	783.511	45.83
Binzhou	3748.5	643.217	17.16
Bozhou	4851	293.575	6.05
Changde	5717.2	515.205	9.01
Changsha	7044.1	1226.816	17.42
Changzhi	3334.6	649.839	19.49
Changzhou	4592	2082.949	45.36
Chaoyang	3044.6	496.376	16.30
Chengde	3473.2	543.004	15.63
Chengdu	14047.6	8084.426	57.55
Chenzhou	4581.8	483.704	10.56
Chifeng	4341.2	652.651	15.03
Chizhou	1403	156.549	11.16
Chongqing	28846.2	7033.574	24.38
Chongzuo	1194.3	186.098	15.58
Chuzhou	3938	485.798	12.34
Dalian	6690.4	4739.327	70.84
Dandong	2444.7	842.188	34.45
Datong	3318.1	694.392	20.93
Daxinganling	511.6	114.424	22.37
Dazhou	5468.1	430.925	7.88
Dezhou	5568.2	699.408	12.56
Dongying	2035.3	534.658	26.27
Erdos	1940.7	438.985	22.62
Fangchenggang	866.9	131.863	15.21
Fushun	2138.1	920.459	43.05
Guangyuan	2484.1	392.252	15.79

Guangzhou	12700.8	12970.73	102.13
Guigang	4118.8	271.472	6.59
Guyuan	1228.2	115.925	9.44
Haikou	2046.1	748.422	36.58
Handan	9174.6	1119.457	12.20
Hangzhou	8700.4	6671.225	76.68
Hefei	5702	2263.288	39.69
Hegang	1058.7	128.822	12.17
Heihe	1633.9	114.306	7.00
Hengshui	4340.8	382.178	8.80
Hengyang	7141.5	850.294	11.91
Hetian	2014.3	181.223	9.00
Heze	8287.8	705.687	8.51
Hezhou	1954.1	178.734	9.15
Hinggan	1613.3	229.829	14.25
League			
Hohhot	2866.6	755.669	26.36
Huai'an	4799.9	924.739	19.27
Huaibei	2114	438.918	20.76
Huaihua	4741.9	402.953	8.50
Huainan	2334	530.365	22.72
Huangshan	1359	215.696	15.87
Huludao	2623.5	660.431	25.17
Hulunbeier	2549.3	708.438	27.79
Jiaxing	4501.7	2526.202	56.12
Jiayuguan	231.9	99.73	43.01
Jilin	4414.7	938.347	21.26
Jinan	6814	3198.532	46.94
Jincheng	2279.2	524.833	23.03
Jinhua	5361.6	1639.551	30.58
Jining	8081.9	1313.794	16.26
Jinzhong	3249.4	644.623	19.84
Jinzhou	3126.4	838.432	26.82
Jixi	1862.1	284.222	15.26
Karamay	391	77.782	19.89
Kashi	3979.3	375.495	9.44
Kiamusze	2552.1	437.396	17.14
Kunming	6432	1715.482	26.67
Laibin	2099.7	194.652	9.27
Laiwu	1298.5	238.67	18.38
Langfang	4358.8	1140.862	26.17
Lanzhou	3616.2	936.616	25.90
Lianyungang	4393.9	1070.188	24.36
Liaocheng	5789.9	695.739	12.02

Liaoyang	1858.8	620.288	33.37
Liaoyuan	1176.6	209.678	17.82
Lijiang	1245	128.848	10.35
Linfen	4316.6	635.71	14.73
Lishui	2117	448.88	21.20
Liu'an	5612	379.776	6.77
Liuzhou	3758.7	1106.096	29.43
Longnan	2567.7	131.365	5.12
Loudi	3785.6	501.615	13.25
Lvliang	3727.1	361.492	9.70
Ma'anshan	1366	516.606	37.82
Maoming	5817.8	620.851	10.67
Mianyang	4613.9	747.878	16.21
Mudanjiang	2798.7	23.65	0.85
Nanchang	5042.6	1235.277	24.50
Nanchong	6278.6	930.867	14.83
Nanjing	8004.7	4714.89	58.90
Nanning	6661.6	1930.216	28.98
Nantong	7282.8	2258.199	31.01
Ningbo	7605.7	7823.121	102.86
Panjin	1392.5	345.817	24.83
Pu'er	2543	233.122	9.17
Qingyang	2211.2	165.467	7.48
Qingyuan	3698.4	1631.227	44.11
Qinhuangdao	2987.6	525.617	17.59
Qinzhou	3079.7	242.058	7.86
Qiqihar	5367	614.791	11.46
Qujing	5855	518.546	8.86
Quzhou	2122.7	805.094	37.93
Sanya	685.4	470.121	68.59
Shangrao	6579.7	468.295	7.12
Shantou	5391	583.743	10.83
Shaoxing	4912.2	1871.129	38.09
Shaoyang	7071.8	521.997	7.38
Shenyang	8106.2	3397.203	41.91
Shiyan	3340.8	607.559	18.19
Shizuishan	725.5	216.019	29.78
Shuo Zhou	1714.9	226.59	13.21
Siping	3386.3	194.77	5.75
Suzhou	10466	7157.478	68.39
Suzhou	5353	333.895	6.24
Taian	5494.2	1085.073	19.75
Taiyuan	4201.6	2481.991	59.07
Taizhou	4618.6	1540.62	33.36

Taizhou	5968.8	2617.891	43.86
Tangshan	7577.3	1647.621	21.74
Tianjin	12938.2	5684.847	43.94
Tieling	2717.7	648.938	23.88
Tonghua	2325.2	333.637	14.35
Tongliao	3139.2	362.058	11.53
Tongling	724	347.393	47.98
Turpan	622.7	115.743	18.59
Ulanqab	2143.6	288.938	13.48
Weifang	9086.2	2003.457	22.05
Weihai	2804.8	1097.634	39.13
Wenzhou	9122.1	2416.055	26.49
Wuhai	532.9	216.1	40.55
Wuhan	9785.4	4789.449	48.94
Wuhu	2263	828.802	36.62
Wuwei	1815.1	153.369	8.45
Wuxi	6372.6	3983.419	62.51
Wuzhong	1273.8	169.305	13.29
Xiangtan	2748.6	497.425	18.10
Xiangyang	5500.3	842.67	15.32
Xiaogan	4814.5	467.471	9.71
Xilin Gol	1028	261.29	25.42
League			
Xingtai	7104.1	731.288	10.29
Xining	2208.7	395.352	17.90
Xinzhou	3067.5	417.393	13.61
Xuancheng	2533	438.373	17.31
Ya'an	1507.2	297.842	19.76
Yancheng	7260.2	1400.917	19.30
Yangquan	1368.5	280.312	20.48
Yangzhou	4459.8	1427.845	32.02
Yantai	6968.2	3138.85	45.05
Yibin	4059.7	830.897	20.47
Yichang	4059.7	1112.95	27.41
Yichun	1148.1	346.149	30.15
Yinchuan	1993.1	782.664	39.27
Yingkou	2428.5	875.818	36.06
Yingtian	1124.9	121.118	10.77
Yiyang	4313.1	384.624	8.92
Yongzhou	5180.2	380.187	7.34
Yueyang	5477.9	605.061	11.05
Yulin	5487.4	400.719	7.30
Yuncheng	5134.8	596.231	11.61
Yunfu	2360.1	266.894	11.31

Yuxi	2304	283.435	12.30
Zaozhuang	3729.3	508.935	13.65
Zhangjiajie	1476.5	131.962	8.94
Zhangjiakou	4345.5	846.879	19.49
Zhangye	1199.5	99.579	8.30
Zhaoqing	3918.1	827.695	21.12
Zhaotong	5213	236.61	4.54
Zhenjiang	3113.3	1012.385	32.52
Zhongshan	3120.9	3035.102	97.25
Zhongwei	1080.8	133.203	12.32
Zhoushan	1121.3	424.164	37.83
Zhuzhou	3855.6	665.169	17.25
Zibo	4530.6	1521.334	33.58
Zigong	2678.9	467.664	17.46
Ziyang	3665.1	218.918	5.97

* The number of city residents is according to the 2010 census.

Table S2: Summary statistics on city-level characteristics in 184 Chinese cities.

City	Number of monitoring stations	Annual average PM _{2.5} levels (µg/m ³)	Annual SD of PM _{2.5} concentrations (µg/m ³)	Annual average temperature (°C)	Annual average relative humidity (%)	Missing rate of PM _{2.5} (%)	GDP per capita (¥10,000)	Average age (year)	Smoking rate (%)
An'shan	7	64	43	11	50	0.5	6.03	48	22.9
Anyang	5	91	69	15	59	0.6	2.54	43	23.6
Baicheng	2	59	38	4	56	3.5	2.17	48	27.8
Baise	2	40	23	23	78	1.4	1.63	43	24.2
Baiyin	2	37	23	12	55	1.5	1.82	48	27.6
Baoji	8	61	51	13	66	0.9	2.63	49	27.6
Baoshan	2	28	17	17	67	1	1.04	46	30.6
Baotou	6	49	34	8	55	0.9	9.28	48	29.3
Bayannur	3	41	28	7	45	0	3.64	48	29.3
Bazhong	6	35	27	18	75	0	0.86	44	27.1
Beihai	4	26	20	24	83	1.6	2.58	43	24.2
Bengbu	6	61	34	16	77	3.8	2.01	42	25.3
Benxi	6	49	37	9	60	1	5.03	48	22.9
Binzhou	3	77	48	14	61	1.1	4.14	41	22.1
Bozhou	2	60	39	17	71	1.4	1.06	42	25.3
Changde	5	56	38	18	78	1.1	2.61	42	27.5
Changsha	2	62	40	18	78	0	2.39	42	27.5
Changzhi	6	64	38	11	58	1.1	2.76	38	26.8
Changzhou	9	57	34	17	73	0.9	6.48	43	24.3
Chaoyang	4	42	33	11	52	6.6	2.15	48	22.9
Chengde	5	38	28	10	56	1.5	2.54	43	25
Chengdu	9	64	45	16	82	7.2	3.95	44	27.1
Chenzhou	5	41	26	19	79	0.9	2.36	42	27.5
Chifeng	4	39	32	8	48	0.9	2.49	48	29.3
Chizhou	4	54	33	18	75	3.5	2.14	42	25.3
Chongqing	17	56	35	19	77	3.2	2.74	45	28.4
Chongzuo	2	33	21	23	77	0	1.95	43	24.2
Chuzhou	3	58	31	16	80	0	1.77	42	25.3
Dalian	10	43	35	13	61	0	7.71	48	22.9
Dandong	4	39	25	11	69	0	2.98	48	22.9
Datong	7	38	28	8	56	3.9	2.09	38	26.8
Daxinganli ng	1	21	20	2	67	1.1	1.92	52	26
Dazhou	5	55	38	18	77	0	1.5	44	27.1
Dezhou	3	89	61	14	67	0	2.98	41	22.1
Dongying	4	68	48	15	61	1	11.6	41	22.1
Erdos	5	26	19	8	47	0.9	13.62	48	29.3
Fangcheng	3	28	19	24	81	2	3.69	43	24.2

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Fushun	6	50	36	7	66	1.5	4.16	48	22.9
Guangyuan	4	24	21	16	69	1	1.3	44	27.1
Guangzhou	12	39	21	22	80	0.9	8.35	34	22.7
Guigang	4	40	25	23	82	0	1.28	43	24.2
Guyuan	3	37	20	8	55	1.8	0.85	46	23.5
Haikou	5	23	15	24	82	0	2.89	45	20.9
Handan	4	92	66	15	58	1.7	2.55	43	25
Hangzhou	11	53	30	18	74	0	6.83	39	23.8
Hefei	10	49	31	20	75	0	4.74	42	25.3
Hegang	4	37	32	6	66	0	2.37	52	26
Heihe	3	25	18	3	63	0	1.56	52	26
Hengshui	3	102	62	11	58	0	1.8	43	25
Hengyang	6	51	34	19	75	0	1.99	42	27.5
Hetian	2	112	97	15	36	0	0.5	43	16.2
Heze	3	86	55	15	69	0	1.39	41	22.1
Hezhou	2	37	22	22	79	0	1.94	43	24.2
Hinggan	3	29	21	7	48	0	1.62	48	29.3
League									
Hohhot	8	43	35	7	47	0	6.51	48	29.3
Huai'an	5	55	34	16	77	0	2.8	43	24.3
Huaibei	3	59	36	16	67	0	2.18	42	25.3
Huaihua	5	44	28	18	80	0	1.42	42	27.5
Huainan	7	57	35	17	77	0	2.59	42	25.3
Huangshan	3	28	19	10	81	4.8	2.28	42	25.3
Huludao	4	51	36	11	53	1.6	2.03	48	22.9
Hulunbeier	2	28	18	0	62	1.3	3.66	48	29.3
Jiaxing	4	49	28	17	82	0	5.1	39	23.8
Jiayuguan	3	28	32	9	47	0	7.93	48	27.6
Jilin	8	54	49	5	65	1.3	4.08	48	27.8
Jinan	8	79	48	15	56	1	5.74	41	22.1
Jincheng	6	62	44	11	59	1.1	3.21	38	26.8
Jinhua	4	52	30	19	74	3.1	3.91	39	23.8
Jining	3	74	46	15	70	0	3.15	41	22.1
Jinzhong	5	60	39	12	58	0	2.35	38	26.8
Jinzhou	5	56	39	11	53	1	2.93	48	22.9
Jixi	4	33	30	7	64	3.6	2.25	52	26
Karamay	5	30	17	11	47	3.7	18.16	43	16.2
Kashi	3	143	124	11	46	0	0.94	43	16.2
Kiamusze	5	26	19	8	71	0	2.01	52	26
Kunming	7	29	13	16	70	0	0.67	46	30.6
Laibin	2	43	30	22	79	0	1.83	43	24.2
Laiwu	3	81	48	14	62	0	4.21	41	22.1
Langfang	5	72	64	11	61	0	3.05	43	25

Lanzhou	5	52	26	10	58	0	3.04	48	27.6
Lianyunga ng	7	43	27	17	73	0	2.62	43	24.3
Liaocheng	4	89	56	14	65	0	2.77	41	22.1
Liaoyang	4	51	43	9	61	1.6	3.96	48	22.9
Liaoyuan	2	48	36	7	66	0	3.49	48	27.8
Lijiang	3	15	4	14	61	2.1	1.15	46	30.6
Linfen	7	89	74	14	54	0	2.07	38	26.8
Lishui	3	36	21	19	74	0	0.86	39	23.8
Liu'an	4	50	28	17	76	0	1.2	42	25.3
Liuzhou	6	45	28	22	72	0	3.35	43	24.2
Longnan	3	26	13	14	67	0	0.66	48	27.6
Loudi	5	47	29	18	80	0	1.8	42	27.5
Lyliang	4	62	36	11	60	0	2.27	38	26.8
Ma'anshan	5	56	35	17	77	0	5.94	42	25.3
Maoming	4	38	23	19	85	0	2.56	34	22.7
Mianyang	10	68	41	13	71	1.3	1.34	44	27.1
Mudanjian g	5	35	27	5	65	0	2.79	52	26
Nanchang	9	43	29	19	75	0	4.38	48	25
Nanchong	6	61	35	18	79	0	1.32	44	27.1
Nanjing	9	55	37	17	73	0	6.26	43	24.3
Nanning	8	37	20	21	81	0	2.7	43	24.2
Nantong	5	51	33	16	78	0	4.69	43	24.3
Ningbo	9	41	26	18	76	0	6.74	39	23.8
Panjin	3	46	34	10	60	1	6.65	48	22.9
Pu'er	2	26	18	20	76	1.2	0.97	46	30.6
Qingyang	3	36	25	11	59	1.4	1.62	48	27.6
Qingyuan	4	38	25	22	77	1.5	3.01	34	22.7
Qinhuangd ao	5	49	40	11	68	0	3.11	43	25
Qinzhou	3	34	23	24	81	0	1.64	43	24.2
Qiqihar	5	33	33	9	62	0	1.59	52	26
Qujing	2	31	14	16	70	0	1.72	46	30.6
Quzhou	3	45	26	19	79	0	3.55	39	23.8
Sanya	2	18	13	23	92	0	3.37	45	20.9
Shangrao	4	43	20	19	78	2.6	1.37	48	25
Shantou	6	33	17	23	77	0	2.23	34	22.7
Shaoxing	4	52	30	18	73	1.1	5.67	39	23.8
Shaoyang	5	56	35	18	81	0	1.03	42	27.5
Shenyang	13	61	49	9	60	1.6	6.19	48	22.9
Shiyan	4	50	30	16	70	0	2.21	46	26.3
Shizuishan	4	45	29	11	46	0	4.11	46	23.5
Shuozhou	5	53	35	8	52	0	3.91	38	26.8

Siping	3	52	41	8	63	0	2.33	48	27.8
Suzhou	8	53	32	16	78	1.4	8.6	43	24.3
Suzhou	8	66	41	17	72	1.4	1.21	42	25.3
Taian	3	67	43	7	65	0	3.73	41	22.1
Taiyuan	9	71	39	6	48	0	4.23	38	26.8
Taizhou	3	69	33	17	78	0	4.34	43	24.3
Taizhou	3	39	23	19	78	0	4.05	39	23.8
Tangshan	6	84	59	12	63	0	5.9	43	25
Tianjin	15	71	51	14	57	0	7.04	37	26.8
Tieling	4	52	41	8	61	1.6	2.66	48	22.9
Tonghua	2	43	30	7	67	0	2.7	48	27.8
Tongliao	3	43	33	8	52	0	3.75	48	29.3
Tongling	6	53	31	18	81	0	6.44	42	25.3
Turpan	2	68	48	16	34	1.6	2.98	43	16.2
Ulanqab	2	36	26	6	47	0	2.57	48	29.3
Weifang	7	68	44	14	62	0	3.4	41	22.1
Weihai	3	35	26	14	62	1	6.93	41	22.1
Wenzhou	4	42	21	18	81	0	3.21	39	23.8
Wuhai	3	48	29	11	44	0	7.34	48	29.3
Wuhan	10	67	45	16	80	1.2	5.64	46	26.3
Wuhu	4	56	34	17	78	0	4.9	42	25.3
Wuwei	2	41	32	10	44	5.7	1.26	48	27.6
Wuxi	8	56	31	17	74	1.5	9.04	43	24.3
Wuzhong	3	47	33	11	47	0	1.65	46	23.5
Xiangtan	7	59	38	17	81	0	2.38	42	27.5
Xiangyang	4	68	49	17	71	2.7	2.8	46	26.3
Xiaogan	2	46	28	18	76	0	1.66	46	26.3
Xilin Gol	2	16	13	5	53	0	5.75	48	29.3
League									
Xingtai	4	88	66	17	59	0	1.7	43	25
Xining	4	44	26	7	58	0	2.84	47	26.7
Xuancheng	3	49	29	18	79	0	2.08	42	25.3
Ya'an	4	41	23	16	80	0	1.9	44	27.1
Yancheng	5	48	35	15	79	0	3.12	43	24.3
Yangquan	6	63	43	15	57	0	3.14	38	26.8
Yangzhou	5	57	34	16	73	0	4.95	43	24.3
Yantai	6	43	31	13	64	0	6.25	41	22.1
Yibin	6	58	40	19	77	0	1.95	44	27.1
Yichang	5	71	51	17	77	0	3.81	46	26.3
Yichun	3	23	20	2	68	1.4	1.76	52	26
Yinchuan	9	51	31	9	49	0	3.83	46	23.5
Yingkou	4	44	37	11	63	1.5	4.13	48	22.9
Yingtian	5	44	24	15	82	0	3.05	48	25
Yiyang	5	46	28	17	78	0	1.65	42	27.5

Yizhou	3	59	46	11	48	0	1.42	38	26.8
Yongzhou	5	48	28	19	81	0	1.48	42	27.5
Yueyang	6	52	31	18	79	0	2.81	42	27.5
Yulin	3	38	25	22	81	1.7	1.52	43	24.2
Yuncheng	6	67	57	15	60	0	1.61	38	26.8
Yunfu	3	36	23	24	83	0	1.67	34	22.7
Yuxi	5	26	13	17	71	0	3.2	46	30.6
Zaozhuang	5	79	47	15	64	0	3.65	41	22.1
Zhangjiajie	4	48	31	17	81	0	1.64	42	27.5
Zhangjiako	5	33	25	12	50	0	2.22	43	25
u									
Zhangye	2	35	10	-2	49	0	0.54	48	27.6
Zhaoqing	4	80	49	16	70	0	2.72	34	22.7
Zhaotong	2	32	17	13	74	2.4	0.73	46	30.6
Zhenjiang	5	58	34	17	72	0	6.29	43	24.3
Zhongshan	4	34	21	23	78	0	5.85	34	22.7
Zhongwei	3	43	28	10	50	0	1.57	46	23.5
Zhoushan	3	28	19	17	81	0	5.65	39	23.8
Zhuzhou	7	58	40	18	79	0	3.31	42	27.5
Zibo	6	80	47	14	62	0	6.33	41	22.1
Zigong	5	73	45	17	82	0	2.42	44	27.1
Ziyang	5	44	31	18	80	0	1.8	44	27.1

¥10,000 (£1169; \$1456; €1377).

SD: standard deviation; GDP: gross domestic product; PM_{2.5}: particulate matter with aerodynamic diameter ≤ 2.5 μm .

Table S3. Spearman correlation coefficients among the exposure variables in 184 Chinese cities, 2014–2017.

Variables	PM _{2.5}	SO ₂	NO ₂	CO	O ₃	Temperature	Relative humidity
PM _{2.5}	1.00	0.56*	0.64*	0.61*	-0.02	-0.26*	-0.08*
SO ₂	—	1.00	0.55*	0.52*	-0.09*	-0.38*	-0.33*
NO ₂	—	—	1.00	0.54*	-0.13*	-0.31*	-0.12*
CO	—	—	—	1.00	-0.21*	-0.28*	-0.04*
O ₃	—	—	—	—	1.00	0.53*	-0.23*
Temperature	—	—	—	—	—	1.00	0.26*
Relative humidity	—	—	—	—	—	—	1.00

PM_{2.5}: particulate matter with aerodynamic diameter ≤ 2.5 μm .

* $P < 0.01$

Table S4. National-average percentage change* (with 95% confidence intervals) in daily hospital admissions for cause-specific cardiovascular disease per 10- $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ concentrations on different lag days in 184 Chinese cities, 2014–2017.

	Lag 0			Lag 1			Lag 2			Lag 01			Lag 02		
Cardiovascular disease	0.26%	(0.17%	to	0.07%	(-0.02%	to	0.01%	(-0.07%	to	0.22%	(0.11%	to	0.21%	(0.08%	to
	0.35%)			0.16%)			0.10%)			0.34%)			0.34%)		
Ischaemic heart disease	0.31%	(0.22%	to	0.11%	(0.01%	to	-0.01%	(-0.10%	to	0.31%	(0.19%	to	0.27%	(0.12%	to
	0.40%)			0.21%)			0.08%)			0.43%)			0.41%)		
Heart failure	0.27%	(0.04%	to	0.04%	(-0.25%	to	0.14%	(-0.11%	to	0.20%	(-0.16%	to	0.29%	(-0.06%	to
	0.51%)			0.33%)			0.38%)			0.56%)			0.63%)		
Heart rhythm disturbances	0.29%	(0.12%	to	0.10%	(-0.07%	to	0.06%	(-0.08%	to	0.24%	(0.05%	to	0.25%	(0.04%	to
	0.46%)			0.27%)			0.21%)			0.43%)			0.47%)		
Ischaemic stroke	0.29%	(0.18%	to	0.08%	(-0.05%	to	0.08%	(-0.04%	to	0.27%	(0.13%	to	0.31%	(0.13%	to
	0.40%)			0.21%)			0.20%)			0.42%)			0.48%)		
Haemorrhagic stroke	-0.02%	(-0.23%	to	-0.21%	(-0.43%	to	-0.12%	(-0.34%	to	-0.17%	(-0.43%	to	-0.23%	(-0.53%	to
	0.19%)			0.01%)			0.09%)			0.08%)			0.08%)		

$\text{PM}_{2.5}$: particulate matter with aerodynamic diameter $\leq 2.5 \mu\text{m}$.

*Adjusted for temperature, relative humidity, calendar time, day of week, and public holiday.

Table S5. National-average percentage change* with 95% confidence interval in daily hospital admissions for cardiovascular disease associated with 10 µg/m³ increase in concurrent day PM_{2.5} concentrations (lag 0) in 184 Chinese cities, 2014–2017, classified by tertiles of city-specific annual average co-pollutants levels.

Co-pollutants	Lower tertile	Middle tertile	Upper tertile
SO ₂	0.29% (0.09% to 0.49%)	0.38% (0.21% to 0.56%)	0.17% (0.07% to 0.28%)
NO ₂	0.51% (0.30% to 0.72%)	0.19% (0.01% to 0.38%)	0.26% (0.17% to 0.34%)
CO	0.40% (0.19% to 0.60%)	0.29% (0.13% to 0.45%)	0.17% (0.06% to 0.27%)
O ₃	0.29% (0.08% to 0.50%)	0.27% (0.09% to 0.45%)	0.25% (0.16% to 0.35%)

PM_{2.5}: particulate matter with aerodynamic diameter ≤2.5 µm.

*Adjusted for temperature, relative humidity, calendar time, day of week, and public holiday.

Table S6. National-average percentage change with 95% confidence interval in daily hospital admissions for cardiovascular disease associated with 10 µg/m³ increase in concurrent day (lag 0) SO₂, NO₂, O₃, and 1 mg/m³ increase in CO in 184 Chinese cities, 2014–2017.

Air pollutants	Percentage change (95% confidence interval)*	<i>P</i>
SO ₂	0.92% (0.71% to 1.14%)	<0.001
NO ₂	1.35% (1.10% to 1.60%)	<0.001
CO	2.15% (1.42% to 2.89%)	<0.001
O ₃	0.16% (0.07% to 0.25%)	<0.001

*Adjusted for temperature, relative humidity, calendar time, day of week, and public holiday.

Table S7. Percentage change* with 95% confidence interval in daily hospital admissions for cause-specific cardiovascular disease for categories of concurrent day PM_{2.5} concentrations (lag 0) in 184 cities in China, 2014–2017.

PM _{2.5} (µg/m ³)	Ischaemic heart disease	Heart failure	Heart rhythm disturbances	Ischaemic stroke
≤15	Reference	Reference	Reference	Reference
15–25	1.2% (0 to 2.4%)	0 (-2.9% to 3.0%)	1.5% (-0.5% to 3.5%)	-0.3% (-1.7% to 1.2%)
25–35	2.9% (1.6% to 4.3%)	0.4% (-2.7% to 3.5%)	2.8% (0.6% to 5.0%)	0.8% (-1.2% to 2.9%)
35–75	3.3% (1.9% to 4.7%)	2.4% (-0.6% to 5.5%)	3.9% (1.6% to 6.2%)	2.1% (0.3% to 4.0%)
>75	5.1% (3.3% to 6.9%)	2.9% (-0.8% to 6.7%)	5.4% (2.3% to 8.7%)	3.1% (0.6% to 5.6%)

PM_{2.5}: particulate matter with aerodynamic diameter ≤2.5 µm.

* Adjusted for temperature, relative humidity, calendar time, day of week, and public holiday.

Table S8. Regional-average percentage change with 95% confidence interval in daily hospital admissions for cardiovascular disease associated with 10 $\mu\text{g}/\text{m}^3$ increase in concurrent day $\text{PM}_{2.5}$ (lag 0) concentrations in 184 Chinese cities, 2014–2017.

Region	Percentage change (95% confidence interval)*	<i>P</i>
East	0.40% (0.29% to 0.51%)	<0.001
Middle-south	0.51% (0.23% to 0.78%)	<0.001
Southwest	-0.06% (-0.65% to 0.52%)	0.833
Northwest	0.09% (-0.08% to 0.26%)	0.287
North	0.13% (0.01% to 0.26%)	0.041
Northeast	0.16% (-0.13% to 0.45%)	0.284

$\text{PM}_{2.5}$: particulate matter with aerodynamic diameter $\leq 2.5 \mu\text{m}$.

* Adjusted for temperature, relative humidity, calendar time, day of week, and public holiday.

Table S9. Annual reduction in cardiovascular admissions and hospitalization cost attributable to PM_{2.5} reduction (achieving various regulatory limits) in China.

PM _{2.5} regulatory limits	Annual reduction in cardiovascular admissions (95% CI)	Annual reduction in hospitalization cost (95% CI, (¥1 million)
CAAQS II (35 µg/m ³)	36448 (24441 to 48471)	418.42 (280.58 to 556.45)
CAAQS I (15 µg/m ³)	85270 (57129 to 113494)	978.90 (655.84 to 1302.91)
WHO (10 µg/m ³)	97516 (65320 to 129820)	1119.48 (749.87 to 1490.33)

PM_{2.5}: particulate matter with aerodynamic diameter ≤ 2.5 µm; CAAQS: Chinese Ambient Air Quality Standards; ¥10,000 (£1169; \$1456; €1377).

Note: The number of cardiovascular admissions in China in 2017 is 9344721, and the average per capita hospitalization cost is ¥11,480, obtained from China Health and Family Planning Yearbook 2017.

Table S10. Results of sensitivity analyses on the association between concurrent day PM_{2.5} concentrations (lag 0) and cardiovascular admissions in 184 Chinese cities, 2014–2017.

Variables	Percentage change (95% confidence interval)	<i>P</i>
Degree of freedom for calendar time		
6	0.26% (0.18% to 0.35%)	< 0.001
7	0.26% (0.17% to 0.35%)	< 0.001
8	0.24% (0.15% to 0.32%)	< 0.001
9	0.25% (0.17% to 0.34%)	< 0.001
10	0.28% (0.20% to 0.36%)	0.001
Degree of freedom for temperature		
2	0.26% (0.17% to 0.34%)	< 0.001
3	0.25% (0.16% to 0.33%)	< 0.001
4	0.26% (0.18% to 0.35%)	< 0.001
5	0.27% (0.18% to 0.35%)	< 0.001
6	0.26% (0.17% to 0.35%)	< 0.001
Degree of freedom for relative humidity		
2	0.27% (0.18% to 0.35%)	< 0.001
3	0.26% (0.17% to 0.35%)	< 0.001
4	0.26% (0.18% to 0.35%)	< 0.001
5	0.26% (0.17% to 0.35%)	< 0.001
6	0.26% (0.17% to 0.35%)	< 0.001
Dataset		
78 cities with 3-year	0.35% (0.15% to 0.55%)	< 0.001
106 cities with 4-year data	0.23% (0.14% to 0.32%)	< 0.001
Spline function		
Penalized spline function	0.26% (0.18% to 0.35%)	< 0.001
Poisson regression	0.25% (0.17% to 0.34%)	< 0.001
Exclusion of cities with ≤ 2 monitors	0.27% (0.18% to 0.36%)	< 0.001
Exposure metrics		
Maximum daily 1-hour	0.17% (0.07% to 0.27%)	0.004
Minimum daily 1-hour	0.28% (0.15% to 0.40%)	< 0.001
Daily 25 percentile	0.19% (0.08% to 0.31%)	< 0.001
Daily median	0.16% (0.06% to 0.27%)	0.007
Daily 75 percentile	0.13% (0.03% to 0.23%)	0.01

PM_{2.5}: particulate matter with aerodynamic diameter ≤ 2.5 μm .



Figure S1. The locations of 184 cities analyzed in this study.

The most populous city in each province was marked with name.

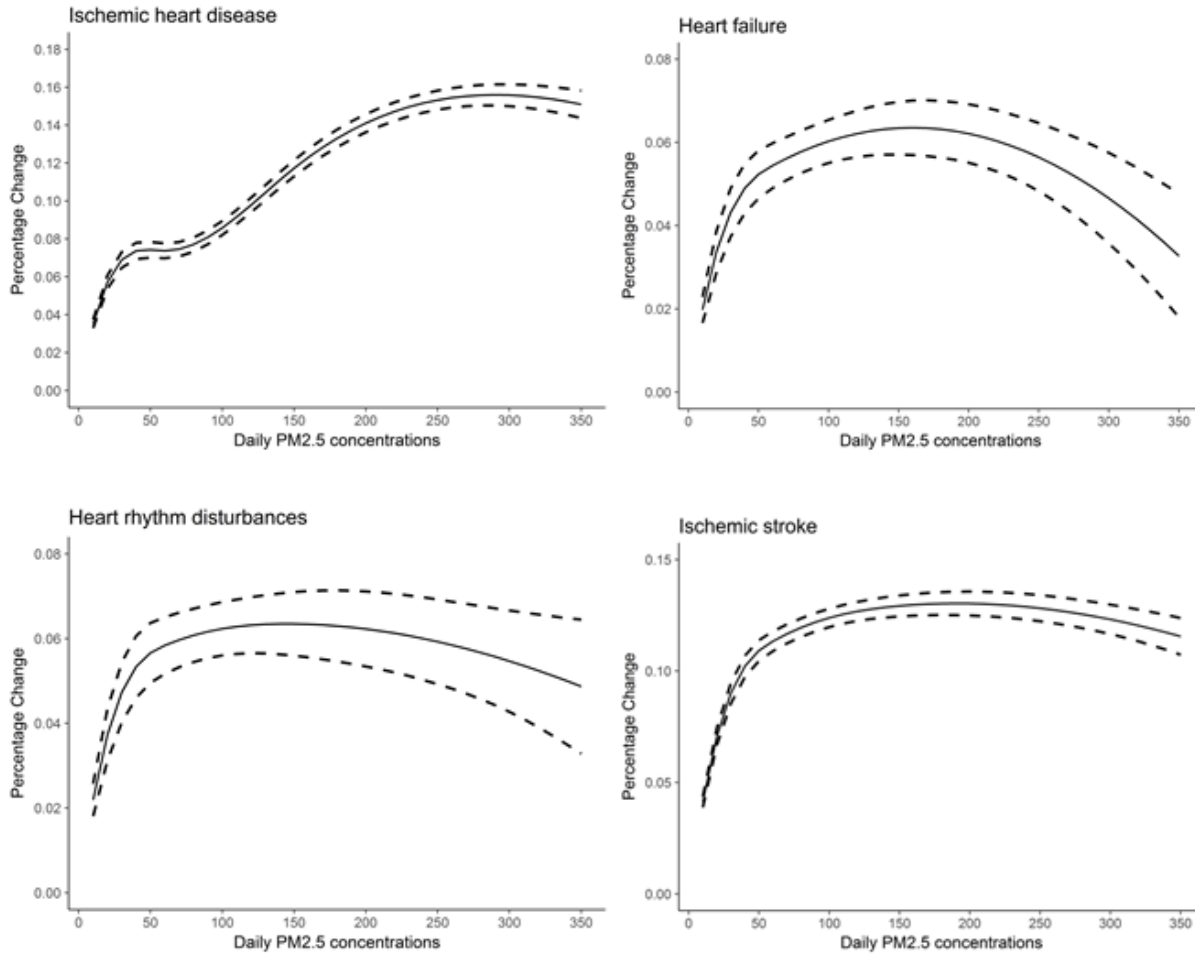


Figure S2. National-average exposure-response association curves between concurrent day PM_{2.5} concentrations (lag 0) and percentage change (%) in daily hospital admissions for cause-specific cardiovascular diseases in 184 Chinese cities, 2014–2017. The solid line represents percentage changes and the dashed lines represent the 95% confidence intervals. PM_{2.5}: particulate matter with aerodynamic diameter ≤ 2.5 μm .